(Please write your Enrolment No. Immediately) *Enrollment No:* .....

**Mid-Term Examination – 2025**

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

**Paper Code:** BS-202 **Subject:** Probability, statistics, and linear prog

**Time:** 1½Hrs. **Max. Marks:** 30

**Note:** Attempt Q. No. 1 which is compulsory and any two more questions from the remaining.

| Q. No. | Question | Marks | CO |
|---|---|---|---|
| 1(a) | Find the binomial distribution for which mean = 2(variance) and mean + variance = 3, and find P ($X \leq 3$). | [2.5] | CO1 |
| 1(b) | A manufacturer produces medicine bottles and finds that 0.1% of bottles are defective. The bottles are packed in boxes containing 500 bottles. Find that in 100 such boxes are expected to contain no defective bottles. | [2.5] | CO1 |
| 1(c) | Find k so that $f(x,y) = kxy$, $1 \leq x \leq y \leq 2$ will be a joint probability density function. | [2.5] | CO2 |
| 1(d) | Calculate the covariance of the following pairs of observations of the variables X and Y (15,44), (20,43), (25,45), (30,37), (40,34), (50, 37). | [2.5] | CO2 |
| 2(a) | The probability density function of a variate X is<br>X : 0 1 2 3 4 5 6<br>P(X): K 3K 5K 7K 9K 11K 13K<br>Find (i) P(X<4) (ii) $P(3 < X \leq 6)$ (iii) What will be the minimum value of k so that $P(X \leq 2) > 3$. | [ 5] | CO1 |
| 2(b) | There are three bags: the First contains 1 white, 2 red, and 3 black balls, the second contains 2 white,3 red, and 1 black balls, and the third contains 3 white,1 red, and 2 black balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from the second bag. | [5] | CO1 |
| 3(a) | A sample of 100 dry battery cells was tested to find the length of life produced the following result, $\mu$ = 12 hours and variance = 3 hours. If the data is normally distributed, what percentage of battery cells are expected to have a life of (i) more than 15 hours (ii) less than 6 hours (iii) between 10 and 14 hours? (Given that P(0<z<1) = 0.3413, P(0<z<0.67) = 0.2487,$P(0 < z < 2) = 0.4772$) | [5] | CO1 |
| 3(b) | Find the coefficient of correlation for 6 college students who got the following percentage of marks in Economics and Statistics:<br>Economics: 10, 14, 18, 22, 26, 30<br>Statistics: 18, 12, 24, 6, 30, 36 | [5] | CO2 |
| 4(a) | **(a)** Three fair of coins are tossed. Let X denote the number of heads on the first two coins, Let Y denote the number of heads on the last two coins. Then (i) Find the joint distribution of X and Y.<br>(ii) Find E[Y\|X=1]. | [5] | CO2 |
| 4(b) | **(b)** The probability density function of a continuous bivariate distribution is given by<br>F (x, y) = $\begin{cases} x + y, \text{where } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, otherwise. \end{cases}$<br>(i) Find the marginal distributions(ii)Find the correlation coefficient of X and Y. | [5] | CO2 |

(Please write your Enrolment No. immediately) Enrolment No.....

**SUPPLEMENTARY MID TERM EXAMINATION**
**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)**
**Fourth Semester, March 2025**

**Paper Code: BS-202** **Subject: Probability, Statistics and Linear Programming**
**Time: $1\frac{1}{2}$ Hrs.** **Max. Marks: 30**

**Note: Attempt O.No.1, which is compulsory, and any two more questions from remaining.**

| Q. No. | Question | Max. Marks | CO(s) |
|---|---|---|---|
| 1(a) | A problem in mathematics is given to three students $A$, $B$, and $C$ whose chance of solving it are 0.5, 0.3 and 0.25 respectively. What is the probability that the problem will be solved? | 2.5 | CO1 |
| 1(b) | In a large corporate computer network, user log-ons to the system can be modelled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes? | 2.5 | CO1 |
| 1(c) | State the Central Limit Theorem. Also, find the standard deviation of the sampling distribution if the sample size is 36 and the population standard deviation is 12? | 2.5 | CO2 |
| 1(d) | A random variable $X$ has the probability mass function $p(x) = \frac{2}{3^x}, x = 1,2,3, \ldots .$ Find its moment generating function about origin. | 2.5 | CO2 |
| 2(a) | Determine the value of $k$ and $E(X^2)$ if the probability density function of a random variable $X$ is given by $f(x) = \begin{cases} 2k & ; 0 \le x < 2 \\ -kx + 4k & ; 2 \le x < 4 \\ 0 & ; \text{otherwise} \end{cases}$ . | 5 | CO1 |
| 2(b) | In a normal distribution, 8% of the items are below 40, and 85% are below 70. Find the mean and standard deviation. Given that: $P(Z < 1.41) = 0.92$ and $P(Z < 1.04) = 0.85$ | 5 | CO1 |
| 3(a) | Find the value of $c$ such that $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < x$ is a joint probability density function of two random variables $X$ and $Y$. Also, find Conditional probability distribution of $X$ given $Y = 1$. | 5 | CO2 |
| 3(b) | Find the covariance for the following joint probability distribution: | 5 | CO2 |
| 4(a) | Consider the completion time of a large commercial development. The proportion of the maximum allowed time to complete a task is modelled as a beta random variable with $\alpha = 2.5$ and $\beta = 1$. What is the probability that the proportion of the maximum time exceeds 0.7? Also, find the mean of the random variable. | 5 | CO1 |
| 4(b) | Suppose that $X$ is a random variable with mean $\mu$ and variance $\sigma^2$. Let $X_1, X_2, \ldots . X_n$ be a random sample of size $n$ from the population represented by $X$. Show that sample variance $S^2$ is an unbiased estimator of population variance $\sigma^2$ . | 5 | CO2 |

| $x$ | 1 | 1 | 2 | 4 |
|---|---|---|---|---|
| $y$ | 3 | 4 | 5 | 6 |
| f($x$,$y$) | 1/8 | 1/4 | 1/2 | 1/8 |

(Please write your Enrolment No. Immediately) *Enrollment No:* .....

**Mid-Term Examination – 2025**

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

**Paper Code:** BS-202 **Subject:** Probability, statistics, and linear prog

**Time:** 1½Hrs. **Max. Marks:** 30

**Note:** Attempt Q. No. 1 which is compulsory and any two more questions from the remaining.

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|---|---|---|---|
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| 1(b) | A manufacturer produces medicine bottles and finds that 0.1% of bottles are defective. The bottles are packed in boxes containing 500 bottles. Find that in 100 such boxes are expected to contain no defective bottles. | [2.5] | CO1 |
| 1(c) | Find k so that $f(x,y) = kxy$, $1 \leq x \leq y \leq 2$ will be a joint probability density function. | [2.5] | CO2 |
| 1(d) | Calculate the covariance of the following pairs of observations of the variables X and Y (15,44), (20,43), (25,45), (30,37), (40,34), (50, 37). | [2.5] | CO2 |
| 2(a) | The probability density function of a variate X is<br>X : 0 1 2 3 4 5 6<br>P(X): K 3K 5K 7K 9K 11K 13K<br>Find (i) P(X<4) (ii) P($3 < X \leq 6$) (iii) What will be the minimum value of k so that P($X \leq 2$) > 3. | [ 5] | CO1 |
| 2(b) | There are three bags: the First contains 1 white, 2 red, and 3 black balls, the second contains 2 white,3 red, and 1 black balls, and the third contains 3 white,1 red, and 2 black balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from the second bag. | [5] | CO1 |
| 3(a) | A sample of 100 dry battery cells was tested to find the length of life produced the following result, $\mu$ = 12 hours and variance = 3 hours. If the data is normally distributed, what percentage of battery cells are expected to have a life of (i) more than 15 hours (ii) less than 6 hours (iii) between 10 and 14 hours? (Given that P(0<z<1) = 0.3413, P(0<z<0.67) = 0.2487,$P(0 < z < 2) = 0.4772$) | [5] | CO1 |
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| 4(a) | **(a)** Three fair of coins are tossed. Let X denote the number of heads on the first two coins, Let Y denote the number of heads on the last two coins. Then (i) Find the joint distribution of X and Y.<br>(ii) Find E[Y\|X=1]. | [5] | CO2 |
| 4(b) | **(b)** The probability density function of a continuous bivariate distribution is given by<br>F (x, y) = $\begin{cases} x + y, \text{where } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, otherwise. \end{cases}$<br>(i) Find the marginal distributions(ii)Find the correlation coefficient of X and Y. | [5] | CO2 |

(Please write your Enrolment No. immediately) Enrolment No.....

**SUPPLEMENTARY MID TERM EXAMINATION**
**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)**
**Fourth Semester, March 2025**

**Paper Code: BS-202** **Subject: Probability, Statistics and Linear Programming**
**Time: $1\frac{1}{2}$ Hrs.** **Max. Marks: 30**

**Note: Attempt O.No.1, which is compulsory, and any two more questions from remaining.**

| Q. No. | Question | Max. Marks | CO(s) |
|---|---|---|---|
| 1(a) | A problem in mathematics is given to three students $A$, $B$, and $C$ whose chance of solving it are 0.5, 0.3 and 0.25 respectively. What is the probability that the problem will be solved? | 2.5 | CO1 |
| 1(b) | In a large corporate computer network, user log-ons to the system can be modelled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes? | 2.5 | CO1 |
| 1(c) | State the Central Limit Theorem. Also, find the standard deviation of the sampling distribution if the sample size is 36 and the population standard deviation is 12? | 2.5 | CO2 |
| 1(d) | A random variable $X$ has the probability mass function $p(x) = \frac{2}{3^x}, x = 1,2,3, \ldots .$ Find its moment generating function about origin. | 2.5 | CO2 |
| 2(a) | Determine the value of $k$ and $E(X^2)$ if the probability density function of a random variable $X$ is given by $f(x) = \begin{cases} 2k & ; 0 \le x < 2 \\ -kx + 4k & ; 2 \le x < 4 \\ 0 & ; \text{otherwise} \end{cases}$ . | 5 | CO1 |
| 2(b) | In a normal distribution, 8% of the items are below 40, and 85% are below 70. Find the mean and standard deviation. Given that: $P(Z < 1.41) = 0.92$ and $P(Z < 1.04) = 0.85$ | 5 | CO1 |
| 3(a) | Find the value of $c$ such that $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < x$ is a joint probability density function of two random variables $X$ and $Y$. Also, find Conditional probability distribution of $X$ given $Y = 1$. | 5 | CO2 |
| 3(b) | Find the covariance for the following joint probability distribution: | 5 | CO2 |
| 4(a) | Consider the completion time of a large commercial development. The proportion of the maximum allowed time to complete a task is modelled as a beta random variable with $\alpha = 2.5$ and $\beta = 1$. What is the probability that the proportion of the maximum time exceeds 0.7? Also, find the mean of the random variable. | 5 | CO1 |
| 4(b) | Suppose that $X$ is a random variable with mean $\mu$ and variance $\sigma^2$. Let $X_1, X_2, \ldots . X_n$ be a random sample of size $n$ from the population represented by $X$. Show that sample variance $S^2$ is an unbiased estimator of population variance $\sigma^2$ . | 5 | CO2 |

| $x$ | 1 | 1 | 2 | 4 |
|---|---|---|---|---|
| $y$ | 3 | 4 | 5 | 6 |
| f($x$,$y$) | 1/8 | 1/4 | 1/2 | 1/8 |

(Please write your Enrolment No. immediately) Enrolment No. _____

# MID TERM EXAMINATION

## B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

**4th Semester, April, 2024**

**Paper Code: BS-202** **Subject: Probability, Statistics and Linear Programming**
**Time: 1½Hrs.** **Max. Marks: 30**

**Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.**

| Q.No. | Questions | Max. Marks | CO(s) |
|---|---|---|---|
| 1(a) | Two bolts are drawn from a box containing 6 good and 8 defective bolts. Find the probability that the second bolt is good if the first one is found to be defective. | 2.5 | CO1 |
| 1(b) | Find the value of $k$ and mean value of $X$. If $X$ is continuous random variable with probability density function given by $f(x)=kx \ (0\le x\le 2)$ $=2k \ (2\le x\le 4)$ | 2.5 | CO1 |
| 1(c) | If $X$ and $Y$ are jointly distributed random variables, then prove that $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$ | 2.5 | CO2 |
| 1(d) | What is the Central Limit theorem & why it is useful? | 2.5 | CO2 |
| 2(a) | A variate X has the probability distribution function<br>x: -3 6 0<br>P(x): 1/6 1/2 1/3<br>Then find the expectation E(x) and $E(x^2)$, hence evaluate $E(2x+1)^2$. | 5.0 | CO1 |
| 2(b) | In a sample of 1000 cases, the mean of a certain test is 14 and deviation is 2.5. Assuming the distribution to be normal, find<br>(i) How many students score between 12 and 15?<br>(ii) How many score below 8?<br>(iii) How many score above 16? | 5.0 | CO1 |
| 3(a) | The Join Probability distribution of two random variables X and Y is given by $P(X=0,Y=1)=1/3$, $P(X=1,Y=-1)=1/3$ and $P(X=1,Y=1)=1/3$<br>Find (i) the marginal distributions of $X$ and $Y$.<br>(ii) the conditional probabilities distribution of $X$, given by $Y=1$. | 5.0 | CO2 |
| 3(b) | Find the moment generating function of the exponential distribution $f(x)=\frac{1}{c}e^{-x/c}$, $0\le x\le\infty, t>0$, hence find its mean and Standard deviation if $\lvert t\rvert<\frac{1}{c}$. | 5.0 | CO2 |
| 4(a) | An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year? (given that $e^{-0.1}= 0.9048$) | 5.0 | CO1 |
| 4(b) | X is the variate with mean 30 and Standard deviation 5, then find the Probability for (i) $26\le X\le 40$ (ii) $\lvert X-30\rvert>5$ | 5.0 | CO2 |

objective?

| | Course 1 | Course 2 | Course 3 | Course 4 |
|---|---|---|---|---|
| Prof 1 | 60 | 40 | 60 | 70 |
| Prof 2 | 20 | 60 | 50 | 70 |
| Prof 3 | 20 | 30 | 40 | 60 |
| TA | 30 | 10 | 20 | 40 |

b) Solve the following LPP using Simplex algorithm.

Max Z = -2x +3y
subject to
$x \leq 5$
$2x-3y \leq 6$
$x, y \geq 0$

**Q9.** a) Find the initial basic feasible solution of the following transportation problem using Vogel's Approximation method. **[7+8=15]**

| | I | II | III | IV | Supply |
|---|---|---|---|---|---|
| A | 21 | 16 | 25 | 13 | 11 |
| B | 17 | 18 | 14 | 23 | 13 |
| C | 32 | 27 | 18 | 41 | 19 |
| Demand | 6 | 10 | 12 | 15 | |

b) Three food products are available at costs of Rs. 10, Rs. 36 and Rs. 24 per unit, respectively. They contain 1,000, 4,000 and 2,000 calories per unit, respectively and 200, 900 and 500 protein units per unit, respectively. It is required to find the minimum-cost diet containing at least 20,000 calories and 3,000 units of protein. Formulate and solve the given problem as an LP problem. Write the dual and using the final simplex table of primal problem, write the solution to the dual problem.

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(Please write your Exam Roll No.) Exam Roll No. .....

# END TERM EXAMINATION

**FOURTH SEMESTER [B.TECH] JUNE 2024**

***Paper Code: BS-202*** ***Subject: Probability, Statistics & Linear Programming***

***Time: 3 Hours*** ***Maximum Marks: 75***

***Note: Attempt five questionin all including Q.No1 which is compulsory. Select one question from each unit. Assume missing data if any. (Scientific Calculators are allowed).***

**Q1.** Attempt all Questions:- **3**

a) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. If two parts are selected at random and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

b) State the Central Limit Theorem. **2**

c) Find the moment-generating function of a random variable X following an exponential distribution. **3**

d) Explain the general procedure for Hypothesis Testing. **2**

e) Solve the following LPP graphically. **5**

$$Min\ Z = -3x + 2y$$
subject to
$$x + y \leq 5$$
$$4 \geq x \geq 0, 6 \geq y \geq 1$$

**UNIT I**

**Q2.** a) Let the random variable $\boldsymbol{X}$ denote the current measured in a thin copper wire in milliamperes. Assume that the range of $\boldsymbol{X}$ is [0, 20 mA], and assume that the probability density function of $\boldsymbol{X}$ is $f(x) = 0.05, 0 \leq x \leq 20$. What is the probability that a measurement of current is between 5 and 10 milliamperes? Also, determine the mean and standard deviation of $\boldsymbol{X}$. **[7+8=15]**

b) Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

i. If a new design attains a good review, then what is the probability that it will be a highly successful product?

b) Three jobs A, B, and C are to be assigned to three machines U, V, and W. The processing cost for each job machine combination is shown in the matrix given below. Determine the allocation that minimizes the overall processing cost. (Cost is in Rs. per unit). **(7)**

| JOB | MACHINE | | |
|---|---|---|---|
| | U | V | W |
| A | 17 | 25 | 31 |
| B | 10 | 25 | 16 |
| C | 12 | 14 | 11 |

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# END TERM EXAMINATION

**FOURTH SEMESTER [B.TECH] JULY 2023**

**Paper Code: BS-202** **Subject: Probability, Statistics and Linear Programming**

**Time: 3 Hours** **Maximum Marks: 75**

***Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.***

Q1 Attempt all questions:

a) Three urns contain 6 red, 4 black; 4 red, 6 black, and 5 red, 5 black balls respectively. One of the urns is selected and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first urn. **(3)**

b) A manufacturer produces two types of models $M_1$ and $M_2$. Each $M_1$ model requires 4 hours of grinding and 2 hours of polishing; whereas $M_2$ model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on an $M_1$ model is ₹ 3 and on an $M_2$ model is ₹ 4. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week. Formulate this as a linear programming problem. **(3)**

c) Determine the value of c that makes the function $f(x,y) = c(x+y)$, a joint probability mass function over points $x = 1,2,3$ and $y = 1,2,3$. Find $E(X)$ and $E(Y)$. **(3)**

d) Find the line of regression or line of best fit for the data given below: $n = 18, \sum x = 12, \sum y = 18, \sum x^2 = 60, \sum y^2 = 96, \sum xy = 48$. **(3)**

e) Find the moment generating function for the distribution defined by $f(x) = \frac{1}{2^x}$, $x = 1,2,3,\ldots\infty$ **(3)**

## UNIT-I

Q2 a) Show that Poisson distribution is a limiting case of Binomial distribution when n is very large and p is small such that np is fixed. If $X$ is a Poisson variate such that $P(X=2) = 9\,P(X=4) + 90\,P(X=6)$. Find the mean and standard deviation of $X$. **(8)**

b) In the normal distribution, 31% of the items are under 45 and 8% are over 64. Determine the mean and variance of the distribution? (Given that $P(Z < -1.4) = 0.08$ and $P(Z > 0.5) = 0.31$). **(7)**

**P.T.O.**